

where organisations should focus their attention, effort, and resources. The metaverse market is still at the foot of its growth curve and the associated technologies are still in the formative stages. As such, some of the key building blocks are still works in progress.

Much of this work is being conducted in open source communities and projects, and represents a fabulous opportunity for open source oriented organisations to ‘get in on the ground floor’ and cement their position as thought leaders and pioneers in this exciting and emerging technology.

Metaverse components and open source

Now let us consider some of the key components that will facilitate the ability and scalability of working in the metaverse.

Identity: The metaverse and its future are intrinsically linked to web3, billed as the next iteration of the World Wide Web. It is based on blockchain technology, which incorporates concepts such as decentralisation and token-based economics. The key term here is ‘decentralised’, i.e., not managed or controlled by a centralised body, or company.

In the present iteration, web2, the market and our data (including our digital identity) is monopolised and monetised by a handful of huge technology companies such as Meta (previously Facebook), Google, Microsoft, etc.

A key tenet of web3 is to liberate, or democratise and decentralise our personal data, including our digital identity. There will, at least in a first phase, not be one single, vast metaverse, but thousands of metaverses.

In the future, some predict that, as there is one internet, there will be one metaverse, interconnecting multiple sub-metaverses in one coherent, standards based, experience. Some metaverses will be private, some will be public, and many will be a combination of the

two. This will lead to a highly transient experience, both as consumers and increasingly as workers. And, therefore, SSI is such an exciting and important facilitation component for the metaverse.

SSI will handle standard authentication requirements, but so much besides. It has the potential to support the concept of ‘living resumes’, which will accurately and exhaustively capture your work, education, and deliverable history. Background verification checks would be made near instantaneous, as all this information could be kept up-to-date and validated through SSI.

Another potential application is pensions, pension contributions and employment benefits, something that will prove increasingly difficult to track as employment tenures continue to shorten.

There is an incredibly significant level of contribution being made by the open source community to SSI. Notable examples are:

- The European Self Sovereign Identity Framework Lab (eSSIF-Lab)
- Hyperledger Foundation (under the Linux Foundation)
- Trust Over IP Foundation (under the Linux Foundation)
- IDUnion
- Sovrin Foundation
- World Wide Web Consortium (W3C)
- Decentralised Identity Foundation (DIF)
- Network of Networks

Smart contracts: Another fundamental application of blockchain technologies that will be a critical component of the metaverse is smart contracts.

There are numerous potential applications of smart contracts that will help resolve usual challenges and areas of contractual inefficiency -- for example, statements of work and payment of workers. Another potentially interesting application is

in the eRFx procurement technology market. In procurement technology, eRFx is an acronym for Electronic Request For [x], where x can be Proposal (RFP), Quotation (RFQ), Information (RFI) or Tender (RFT). A huge amount of investment in the typical enterprise is communicated and tendered through this procurement process. By leveraging smart contracts and linking this to the new talent delivery model we can significantly streamline the entire process.

As an application, the smart contract lends itself much more readily to the ‘matching’ process. It also provides invaluable information to ‘shift-left’ the visibility of demand skills to the talent community and vendors alike. Fulfilment and payment milestones could be automated because of smart contracts, to name just a few benefits.

Another entity associated with smart contracts is a decentralised autonomous organisation (DAO). A DAO is effectively a group of people who agree to abide by certain rules for a common purpose. Those rules are written into the code of the organisation via smart contracts on the blockchain (algorithms that run when certain criteria are met). Open source projects and their communities require a similar level of coordination and governance.

Given that the next iteration of the internet, web3, is built upon blockchain and is intrinsically linked to the metaverse, there is a potential opportunity to rethink where code is developed and versioned, as well as how to foster collaboration, provide training, and support communities of practice in the open source community. DAOs could be a vehicle to provide the governance layer for that vision.

Digital twins

For the metaverse to become a genuine place for work, it will be necessary to replicate assets (‘assets’ being anything from a container through to an entire operating system and everything in